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RESEARCH ARTICLE OPEN ACCESS

International Evidence on the Relationship Between Financial Literacy and Corporate Cash Holdings

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ABSTRACT

This study is the first investigation of the relationship between corporate cash holdings and financial literacy (FL). We find that a one-standard-deviation increase in FL is associated with an 8%–27% reduction in corporate cash holdings, depending on the measure of FL used. Moreover, higher FL reduces the market value of excess cash, indicating that financially literate environments discourage value-destroying cash accumulation. The results remain robust across alternative definitions of FL and cash holdings, estimation methods, and country subsamples. We further rule out the influence of endogeneity, refinancing risk, and variation in cash needs. Additional analyses identify agency costs as the primary channel through which FL affects corporate liquidity policies. The evidence suggests that financial literacy mitigates the agency motive for holding cash and serves as an effective substitute for monitoring opportunistic firm behavior when formal country governance structures are weak.

1 | Introduction

Financial literacy (FL) enhances individuals' ability to process economic information and make informed financial decisions. Prior studies have focused on the positive effects of FL as a tool for improving personal finances, such as retirement planning (Lusardi and Mitchell 2014), wealth accumulation (Stango and Zinman 2009), and stock market participation (Van Rooij et al. 2011).

Few studies have explored the relationship between FL and firm behavior. Jin et al. (2021) find that FL mitigates opportunistic bank loan loss provisions. More recently, Jia et al. (2024) provide evidence that FL reduces information friction in capital markets, which consequently mitigates IPO underpricing. Building on this emerging line of research, we argue that FL reduces information asymmetry in capital markets and enhances firm transparency; thus, FL serves as an informal mechanism that monitors firm

behavior. This study investigates the relationship between FL and corporate cash holdings.

A particularly influential perspective regarding the motives for firms to hold cash stems from agency theory. Jensen's (1986) free cash flow hypothesis posits that managers have incentives to retain excess liquidity to increase discretionary control over resources, potentially at the expense of shareholder value. This idea has been empirically supported by cross-sectional studies showing that firms with weaker governance hold more cash, consistent with the view that agency conflicts exacerbate the accumulation of liquid assets (Dittmar et al. 2003; Kalcheva and Lins 2007; Harford et al. 2008).

In this study, we argue that FL, by reducing information asymmetry in capital markets and enhancing firm transparency, serves as an informal monitoring mechanism that influences corporate cash holdings. Specifically, we contend that FL mitigates the

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agency motive for holding cash. Accordingly, our first hypothesis is that FL reduces corporate cash holdings.

Scholars have examined the value to shareholders of an additional dollar of corporate cash. The literature well establishes that undisciplined managers may misuse a firm's cash flows to the detriment of shareholders and other stakeholders (Jensen 1986; Lie 2000; Bates 2005). Excess cash is more easily diverted to private benefits or pet projects by managers than are other assets (Lie 2000). Consequently, investors apply a higher discount to cash balances, reducing the firm's market value.

In this study, we posit that FL serves as an informal monitoring mechanism that constrains managerial behavior and thus mitigates the agency costs of excess cash. Accordingly, our second hypothesis is that FL is negatively associated with the market value of corporate cash holdings.

Using a cross-country sample, this study provides evidence of the governance role of FL in corporate cash holdings. Our empirical results support both hypotheses. We document a significant negative relationship between FL and corporate cash holdings. Depending on the measure of FL, a one-standard-deviation increase in FL reduces cash holdings by 8%–27%. Robustness checks confirm this negative association in both developed and developing countries. The relationship also persists across alternative measures of cash holdings. Consistent with FL mitigating the agency motive for holding cash, we find that the negative effect is stronger in firms with more severe agency problems. A key finding is that FL serves as a substitute for weak formal country-level governance. The negative relationship between FL and cash holdings is more pronounced when country governance is weak, and the economic magnitude of FL often exceeds that of aggregate governance quality.

Addressing potential endogeneity, two-stage least squares instrumental variable (2SLS-IV) regressions confirm the reliability of our baseline results. Further analysis of the market value of excess cash supports the monitoring role of FL; specifically, FL significantly reduces the valuation of excess cash. Finally, we rule out confounding effects from refinancing risk or the operational nature of cash needs.

This study makes several contributions to the literature. First, we provide the first investigation in the literature of the relationship between FL and corporate cash holdings. We identify FL in society as a novel determinant of corporate cash holdings. Our results reveal a robust negative relationship between FL and cash holdings across a global sample, introducing FL as a new driver in behavioral corporate finance.

Second, we contribute to the literature on institutional governance by demonstrating that FL serves as an effective informal monitoring mechanism. Our evidence shows that FL constrains self-interested managerial behavior and acts as a substitute for weak formal country-level governance.

Finally, we extend the literature on the market value of corporate cash by showing that FL not only reduces the level of cash

holdings but also lowers the market valuation of excess cash. In environments with higher FL, investors apply a greater discount to excess cash, reflecting enhanced information processing and stronger managerial accountability.

The remainder of the paper is organized as follows. Section 2 reviews related literature and develops the hypotheses. Section 3 describes the research design and methodology. Section 4 presents the sample, empirical results, and robustness tests. Section 5 concludes.

2 | Literature Review and Hypotheses

2.1 | Financial Literacy: Concept and Measurement

FL is broadly defined as the ability to understand and apply financial concepts such as compounding, inflation, risk diversification, and investment returns. Lusardi and Mitchell (2014) provide a fundamental review showing that financial literacy levels are surprisingly low worldwide; only about one-third of adults can answer basic financial questions correctly.

At the global level, Klapper et al. (2015, Klapper and Lusardi, 2020) provide static country-level measures of FL. Their survey, which covers more than 140 economies, shows large cross-country variation across advanced and developing economies. For instance, FL tends to be higher in countries with stronger educational systems and deeper financial markets, while developing economies often lag. Jia et al. (2024) construct a time-varying annual measure of FL by using principal component indices based on public education, tertiary enrollment, nonlife insurance, and the number of bank branches.

An emerging line of research extends FL beyond households to firms and markets. Jin et al. (2021) show that FL mitigates opportunistic behavior in bank loan loss provisioning, suggesting that knowledgeable stakeholders constrain managerial discretion. Jia et al. (2024) document that higher FL reduces IPO underpricing by lowering information frictions, illustrating how FL improves capital market efficiency.

Building on this foundation, FL can be conceptualized as an informal monitoring device. Unlike trust or civic norms, FL operates through knowledge: It equips shareholders to evaluate liquidity policies, creditors to impose covenants against opportunistic hoarding, and analysts to interpret disclosures for broader markets.

2.2 | Hypothesis Development

2.2.1 | FL and the Level of Corporate Cash Holdings

Corporate cash holdings are shaped not only by transaction and precautionary motives (Opler et al. 1999; Bates et al. 2009) but also by agency considerations. According to Jensen's (1986) free cash flow hypothesis, excess liquidity provides managers

with discretionary resources that can be diverted into inefficient projects or retained to shield them from external discipline. Prior studies show that governance quality influences both the level of cash holdings and the marginal value investors assign to them (Dittmar and Mahrt-Smith 2007; Pinkowitz et al. 2006; Harford et al. 2008).

The literature on corporate cash holdings consistently highlights that liquidity policy is highly sensitive to monitoring. We argue that FL acts as an informal governance mechanism. Unlike formal institutions, laws, shareholder rights, or board structures, FL operates through knowledge diffusion at the societal level. Literate stakeholders are more capable of interpreting financial disclosures, distinguishing precautionary from opportunistic liquidity, and disciplining managers accordingly. Shareholders can challenge hoarding through market reactions and voting power, creditors can design stricter covenants, and analysts can produce sharper signals that shape investor perceptions. In this way, FL mitigates the agency motive for excess cash holdings, even where formal institutions are limited. Accordingly, our first hypothesis is as follows:

Hypothesis 1. FL and firm cash holdings are negatively related.

2.2.2 | FL and the Market Value of Corporate Cash

Faulkender and Wang (2006) show that the marginal value of cash is not constant but varies systematically with firm characteristics. They find evidence showing that the value of cash depends on the trade-off between its role as a buffer and the potential for wasteful managerial use.

A related line of research emphasizes governance and monitoring. Dittmar and Mahrt-Smith (2007) demonstrate that in poorly governed firms, the value of an incremental dollar of cash is close to zero, as investors expect cash to be wasted. Conversely, in well-governed firms, each additional dollar is valued at close to one (Pinkowitz et al. 2006; Fresard and Salva 2010).

Most of the literature focuses on formal governance mechanisms, legal protections, shareholder rights, or board structures, while giving less attention to informal or societal mechanisms that might influence investor perceptions.

If FL improves stakeholders' ability to process disclosures and monitor firms, it may serve a role similar to that of governance in shaping the value of cash. Specifically, in financially literate societies, investors are more likely to recognize when cash balances reflect prudent risk management versus managerial opportunism. In this way, FL operates as an informal monitoring mechanism, amplifying the disciplining effect of governance and reducing the agency-related component of cash's value. Thus, our second hypothesis is as follows:

Hypothesis 2. FL reduces the market value of corporate cash holdings.

3 | Regression Models

3.1 | Baseline Model: FL and the Level of Corporate Cash Holdings

We adapt prior models to examine the relationship between FL and corporate cash holdings (Opler et al. 1999; Bates et al. 2009). Our baseline regression specification is as follows:

$$\text{CASH}_{i,j,t} = \beta_0 + \beta_1 \text{FL}_{j,t-1} + \beta_2 \text{Firm controls}_{i,j,t-1} + \beta_p \text{Country controls}_{j,t-1} + \text{Industry FE} + \text{Year FE} + \varepsilon_{i,j,t} \quad (1)$$

where $\text{CASH}_{i,j,t}$ is the natural logarithm of (cash and cash equivalents divided by book value of total assets) for firm i in country j at year t . We measure FL using FLK and FLJ alternately. FLK is a time-invariant country-level indicator derived from survey responses in Klapper et al. (2015). FLJ is a time-varying annual measure constructed as the first principal component extracted via principal component analysis from multiple country-level financial literacy proxies (Jia et al. 2024). Firm-level controls are drawn from the cash holdings literature and include the market-to-book ratio (MTB), firm size ($\ln(\text{TA})$), cash flow (CF), net working capital (NWC), capital expenditures (CAPEX), leverage, cash flow volatility, R&D expenses (RD), acquisition expenditures, and dividend payments. Variable definitions are provided in the Appendix. Country-level controls include macroeconomic and financial development indicators commonly used in cross-country studies: domestic private credit to GDP, stock market capitalization to GDP, nominal GDP growth, and GDP per capita (in constant USD). We also include societal trust (Trust) as a control, given prior evidence that trust facilitates knowledge acquisition and dissemination (Hardwig 1991; Abrams et al. 2003). All specifications include industry and year fixed effects. Standard errors are robust and clustered at the firm level.

3.2 | FL and the Market Value of Corporate Cash

To test Hypothesis 2, we estimate a valuation regression adapted from Fama and French (1998) and Dittmar and Mahrt-Smith (2007). The model is specified as follows:

$$\text{MV}_{i,j,t} = \beta_0 + \beta_1 \text{FL}_{j,t-1} + \beta_2 \text{Xcash}_{i,j,t-1} + \beta_3 (\text{FL}_{j,t-1} \times \text{Xcash}_{i,j,t-1}) + \beta_k \text{Firm controls}_{i,j,t-1} + \beta_p \text{Country controls}_{j,t-1} + \text{Industry FE} + \text{Year FE} + \varepsilon_{i,j,t} \quad (2)$$

where $\text{MV}_{i,j,t}$ is the market value of firm i in country j at time t , computed as the sum of market capitalization and total debt, scaled by total assets (MV/TA), following Pinkowitz et al. (2006) and Bates et al. (2009).

We measure FL using FLK and FLJ alternately. Excess cash ($\text{Xcash}_{i,j,t-1}$) is the residual from the expected cash model of Opler et al. (1999), estimated separately for each country in the sample (Fresard and Salva 2010) and scaled by total assets at time t . The coefficient of primary interest is β_3 , which captures the interaction between $\text{FL}_{j,t-1}$ and $\text{Xcash}_{i,j,t-1}$.

Firm-level controls follow Fama and French (1998) and include earnings before extraordinary items, R&D expense, interest expense, and dividends, all measured at time t , as well as their changes from $t - 2$ to t and from t to $t + 2$. We also include changes in net assets over the same horizons and the market value at $t + 2$. All variables are scaled by total assets at time t . Country-level controls are identical to those in Equation (1). All specifications include industry and year fixed effects. Standard errors are robust to heteroskedasticity and clustered at the firm level.

4 | Sample and Results

4.1 | Sample

We obtain firm-level accounting and market data from the Thomson One Eikon database for the period 2000–2022. To mitigate confounding effects from cross-listing regulations, we exclude American Depositary Receipts (ADRs) and Global Depositary Receipts (GDRs). We also exclude utility and financial firms due to their distinct regulatory environments and cash management practices. After applying these filters and requiring nonmissing data for the baseline regression variables, the final sample comprises 214,270 firm-year observations across 38 countries. All financial variables are winsorized at the 1st and 99th percentiles to reduce the influence of outliers.

4.1.1 | Descriptive Statistics

Table 1 presents descriptive statistics for the final sample of 214,270 firm-year observations. The mean cash-to-assets ratio (Cash/TA) is 0.0906, consistent with prior cross-country evidence and slightly below U.S. levels reported in the early 2000s (Bates et al. 2009).

FL exhibits meaningful variation across countries. The survey-based measure FLK has a mean of 45.3, while the principal component-based FLJ is standardized with a mean near zero and greater dispersion, facilitating interpretation in regression analysis. Other firm characteristics are in line with the cash holdings literature.

4.2 | Results

4.2.1 | Baseline Results

Table 2 reports the baseline regression results testing Hypothesis 1. FL is significantly and negatively associated with corporate cash holdings, after controlling for firm- and country-level determinants, industry fixed effects, and year fixed effects.

Using the survey-based measure FLK, the coefficient is -0.0061 ($p = 0.0039$). Given a standard deviation of 14.19 in FLK, a one-standard-deviation increase in FL reduces cash holdings by 8.3% relative to the sample mean. With the principal component-based FLJ, the coefficient is -0.1871 ($p < 0.0001$), implying a 27.4% reduction in cash holdings for a one-standard-deviation increase. These effects are economically substantial and comparable in magnitude to the governance effects documented by Dittmar et al.

(2003), who find that strong shareholder rights significantly curb cash accumulation.

The firm-level control variables enter with expected signs, consistent with Opler et al. (1999). At the country level, greater private credit availability is associated with higher cash balances, whereas larger equity markets (market capitalization to GDP) correlate with lower cash, supporting the substitutability between internal liquidity and external finance (Pinkowitz et al. 2006).

Collectively, these findings establish societal FL as a potent informal governance mechanism, akin to institutional investor oversight, in constraining managerial cash hoarding.

4.3 | Robustness Checks

4.3.1 | Developed Versus Developing Countries

In Table 3, we document robustness tests that split the sample into developed and developing countries. The negative association between FL and corporate cash holdings persists in both subsamples, reinforcing Hypothesis 1.

In developed markets, the coefficients are -0.0404 ($p < 0.0001$) for FLK and -0.2251 ($p < 0.0001$) for FLJ. In developing markets, the estimates are -0.0134 ($p = 0.0328$) for FLK and -0.1683 ($p = 0.0019$) for FLJ. The effect is statistically significant and economically stronger in developed economies, consistent with Fresard and Salva (2010), who show that a more sophisticated institutional environment enhances the governance impact of external monitoring mechanisms on cash allocation.

4.3.2 | Alternative Measures of Corporate Cash Holdings

We further test the robustness of our findings using alternative measures of cash holdings. The results, reported in Table 4, replace the baseline dependent variable with $\ln(\frac{\text{Cash}}{\text{Net Assets}})$ and $\ln(\text{Cash}/\text{Sales})$. The negative relationship between FL and corporate cash holdings remains robust and, in nearly all specifications, statistically significant. A one-standard-deviation increase in FLJ reduces cash holdings by approximately 27%–30% of the sample mean across these measures, confirming the economic magnitude observed in the baseline model.

4.4 | The Monitoring Channel

Agency conflicts incentivize managers to hoard cash for private benefits rather than shareholder value (Jensen 1986). We hypothesize that FL mitigates this agency-driven cash accumulation by enhancing external monitoring and reducing the opacity that enables managerial opportunism. To test this channel, we estimate an augmented specification that interacts FL with firm-level agency cost proxies (AC):

$$\begin{aligned} \text{CASH}_{i,j,t} = & \beta_0 + \beta_1 \text{FL}_{j,t-1} + \beta_2 \text{AC}_{i,j,t-1} + \beta_3 (\text{FL}_{j,t-1} \times \text{AC}_{i,j,t-1}) \\ & + \beta_k \text{Firm controls}_{i,j,t-1} + \beta_p \text{Country controls}_{j,t-1} \\ & + \text{Industry FE} + \text{Year FE} + \varepsilon_{i,j,t} \end{aligned} \quad (3)$$

TABLE 1 | Descriptive statistics.

Parameter	N	Mean	S.D.	25th percentile	50th percentile	75th percentile
FLK	214,270	45.33	14.19	32	43	57
FLJ	160,260	0.13	1.71	−1.07	−0.66	2.05
Cash/TA	214,270	0.09	4.10	0.05	0.11	0.24
LN(Cash/TA)	214,270	−2.40	1.41	−3.06	−2.17	−1.43
Market-to-book	214,270	1.85	2.84	0.93	1.25	1.95
Ln(TA)	214,270	5.63	2.14	4.26	5.55	6.96
Cash flow/assets	214,270	0.06	0.20	0.04	0.08	0.13
NWC/assets	214,270	0.00	0.53	−0.08	0.01	0.13
Capex/assets	214,270	0.05	0.06	0.01	0.03	0.07
Leverage	214,270	0.25	0.23	0.05	0.20	0.52
Industry sigma	214,270	0.14	1.20	0.02	0.06	0.09
RD/sales	214,270	0.06	0.31	0.00	0.00	0.03
Domestic credit/GDP (%)	214,270	142.66	46.21	120.27	157.78	177.92
Market Cap/GDP(%)	214,270	100.00	41.73	61.47	99.03	132.78
GDP growth (%)	214,270	2.40	3.19	0.93	1.92	3.89
GDP per capita	214,270	32728	20393	9473	34368	52789
Trust	214,270	0.40	0.10	0.36	0.37	0.43

Note: Summary statistics for the final sample of 214,270 firm-year observations across 38 countries over the period 2000–2022. Cash/TA is cash and cash equivalents divided by total assets. FLK is the survey-based static financial literacy measure from Klapper et al. (2015). FLJ is the principal component-based time-varying financial literacy index following Jia et al. (2024). All other variables are as defined in the Appendix. Continuous variables are winsorized at the 1st and 99th percentiles.

We employ three proxies for AC. Free cash flow (FCF), computed as operating cash flow minus capital expenditures and scaled by total assets, measures discretionary resources vulnerable to misuse (Bates 2005). Accruals quality (AQ), defined as the standard deviation of abnormal accruals, captures financial reporting transparency; poorer accruals quality indicates greater potential for earnings manipulation and reduced investor oversight (Lee and Masulis 2009). The sales-to-assets ratio reflects operational efficiency, with lower values signaling agency-induced asset underutilization (Ang et al. 2000). A negative β_3 indicates that FL's dampening effect on cash holdings strengthens with agency risk.

Results are reported in Table 5. The interaction $FLK \times FCF$ is negative and significant at the 1% level, implying that FL significantly curbs cash holdings in high free cash flow firms. The $FLJ \times FCF$ interaction is directionally negative but statistically insignificant. Interactions with accruals quality ($FLK \times AQ$ and $FLJ \times AQ$) are insignificant. However, both $FLK \times Sales/Assets$ and $FLJ \times Sales/Assets$ are negative and significant at the 1% level, indicating that FL constrains cash accumulation in firms with inefficient asset deployment. The evidence provides robust support for the agency channel, particularly through free cash flow and asset utilization mechanisms. These findings demonstrate that societal FL functions as an informal governance tool that disciplines managerial cash policies where agency frictions are most pronounced.

4.5 | Interactions Between FL and Country Governance

To provide further evidence that FL serves as an effective substitute for formal institutional governance, we investigate whether its negative effect on corporate cash holdings is more pronounced in countries with weak governance structures. A country is classified as having weak governance if its score on a given indicator falls below the sample median. We employ six widely used governance metrics as follows: five from the World Bank's Worldwide Governance Indicators, Control of Corruption, Government Effectiveness, Rule of Law, Voice and Accountability, and Regulatory Quality, and the anti-self-dealing index (Djankov et al. 2008) as a direct measure of minority investor protection.

Results are presented in Table 6, with separate panels corresponding to each governance dimension. Consistent with agency theory, weak institutional quality is positively and significantly associated with higher corporate cash holdings across all specifications. For example, in Panel A, firms operating under low Rule of Law or low Voice and Accountability exhibit significantly elevated cash balances ($p < 0.01$), reflecting reduced external discipline and heightened managerial entrenchment.

The coefficient of primary interest is the interaction term $FL \times Low\ Governance$. A negative and significant estimate indicates

TABLE 2 | Baseline regressions of corporate cash holdings on financial literacy.

Parameter	(1)	(2)	(3)
Intercept	−2.2037***	−2.0308***	−2.1490***
FLK		−0.0061***	
FLJ			−0.1871***
Market to book	0.0326***	0.0332***	0.0345***
Firm size	−0.0474***	−0.0467***	−0.0418***
Cash flow/assets	−0.0095	−0.0122	−0.2347
NWC/assets	−0.5234***	−0.5249***	−0.5562***
Capex/assets	−1.8124***	−1.8019***	−2.1156***
Leverage	−2.1780***	−2.1829***	−2.1123***
Industry sigma	−0.0000***	−0.0000***	−0.0000***
R&D/sales	0.3834***	0.3849***	0.3839***
Acquisition activity	−0.6672***	−0.6499***	−0.4964***
Dividend dummy	−0.0211	−0.0239	−0.1622**
Domestic credit/GDP	0.0074***	0.0068***	0.0123***
Market capitalization/GDP	−0.0006***	−0.0009***	−0.0054***
GDP growth	0.0055**	0.0047**	−0.0007
GDP per capita	−0.0000***	−0.0000**	−0.0000
Trust	−0.0384*	−0.0645**	−1.2713***
Industry fixed effect	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Country fixed effects	Yes	No	Yes
Observations	214,270	214,270	160,260
Adjusted R^2	0.2168	0.2172	0.2728

Note: This table reports OLS estimates of Equation (1) examining the relationship between country-level financial literacy and corporate cash holdings. The dependent variable is $CASH_{i,j,t}$, defined as the natural logarithm of (cash and cash equivalents/total assets). FLK is the survey-based financial literacy measure from Klapper et al. (2015). FLJ is the principal component-based financial literacy index following Jia et al. (2024). All independent variables are lagged 1 year. All specifications include firm-level controls, country-level controls, and industry and year fixed effects. Country fixed effects are excluded when FLK is the measure of literacy due to its time-invariant nature. Standard errors are robust and clustered at the firm level. Standard errors are robust and clustered at the firm level.

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

TABLE 3 | Robustness, developed versus developing countries.

Parameter	Developed countries		Developing countries	
	(1)	(2)	(3)	(4)
Intercept	1.0751***	−1.630***	−2.5735***	−3.2654***
FLK	−0.0404***		−0.0134**	
FLJ		−0.2251***		−0.1683***
Firm and country controls	Yes	Yes	Yes	Yes
Industry fixed effect	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	No	Yes	No	Yes
Observations	129,468	101,348	84,802	58,912
Adjusted R^2	0.2277	0.2452	0.3450	0.3837

Note: This table reports OLS estimates of Equation (1) separately for subsamples of developed and developing countries. Firm controls include all the firm-level and country-level control variables in the baseline model. Standard errors are robust and clustered at the firm level.

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

TABLE 4 | Robustness, alternative measures of cash holdings.

Parameter	Cash = ln (Cash/NA)		Cash = ln (Cash/Sales)	
	(1)	(2)	(3)	(4)
Intercept	−1.7541***	−1.8506***	−2.2323***	−2.0844***
FLK	−0.0055**		−0.0006*	
FLJ		−0.2061***		−0.1820***
Firm and country controls	Yes	Yes	Yes	Yes
Industry fixed effect	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	No	Yes	No	Yes
Observations	214,270	160,260	214,270	160,260
Adjusted R ²	0.2428	0.2962	0.1892	0.2481

Note: This table reports OLS estimates of Equation (1) using alternative dependent variables: ln(Cash/Net Assets) and ln(Cash/Sales). Standard errors are robust and clustered at the firm level.

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

TABLE 5 | The agency monitoring channel.

Parameter	FCF	FCF	AQ	AQ	Sales/Assets	Sales/Assets
Intercept	−1.9314***	−2.0727***	−1.8954***	−2.0167***	−1.9768***	−2.1254***
FLK	−0.0063***		−0.0062**		−0.0052***	
FLJ		−0.1888***		−0.1893***		−0.1564***
FLK*FCF	−0.0001***					
FLJ*FCF		−0.0049				
FLK*AQ			0.0035			
FLJ*AQ				0.0772		
FLK*Sales/assets					−0.0008***	
FLJ*Sales/assets						−0.0329***
FCF	−0.0002***	0.0002				
AQ			−0.2370	0.1426***		
Sales/assets					−0.0280**	−0.0146
Firm and country controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	209,252	137,681	185,821	142,494	214,270	160,260
Adjusted R ²	0.2193	0.2720	0.2189	0.2663	0.2285	0.2750

Note: This table reports OLS estimates of Equation (3), testing whether financial literacy (FL) mitigates agency-driven cash holdings. Agency costs are proxied by (i) free cash flow (FCF), defined as operating cash flow minus capital expenditures scaled by total assets; (ii) accruals quality (AQ), measured as the standard deviation of abnormal accruals (Lee and Masulis 2009); and (iii) sales-to-assets ratio (Ang et al. 2000). The interaction term $FL_{j,t-1} \times \text{Agency Cost}_{i,j,t-1}$ captures the differential effect of FL in high-agency-cost firms. All independent variables are lagged 1 year. Specifications include the full set of firm- and country-level controls from Equation (1), along with industry and year fixed effects. Standard errors are robust and clustered at the firm level.

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

that FL compensates for institutional voids by constraining opportunistic cash accumulation. This prediction is strongly supported: Interaction terms are uniformly negative and statistically significant at the 1% level in nearly all cases. Representative coefficients include $FLK \times \text{Low Rule of Law} = -0.0085$ ($p < 0.0001$), $FLJ \times \text{Low Government Effectiveness} = -0.0797$ ($p <$

0.0001), and $FLJ \times \text{Low Control of Corruption} = -0.0688$ ($p < 0.0001$). The final estimate implies that a one-standard-deviation increase in FLJ reduces cash holdings by an additional 11.1% of the sample mean in environments with weak corruption control, over and above the baseline FL effect. These results resonate with Djankov et al. (2008), who document the critical role of

TABLE 6 | Financial literacy as a substitute for weak country-level governance.

Panel A		Rule of law		Voice and accountability	
Parameter	(1)	(2)	(3)	(4)	
Intercept	−2.0432***	−2.1501***	−2.0510***	−2.1556***	
FLK	−0.0057***		−0.0056***		
FLK*Low Rule of law	−0.0085***				
FLK*Low Voice and accountability			−0.0082***		
FLJ		−0.1789***		−0.1834***	
FLJ*Low Rule of law		−0.0807***			
FLJ*Low Voice and accountability				−0.0709***	
Low Rule of Law	0.5329***	0.1442***			
Low Voice and accountability			0.4878***	0.0858***	
Firm and country controls	Yes	Yes	Yes	Yes	
Fixed effects	Yes	Yes	Yes	Yes	
Observations	214,270	160,260	214,270	160,260	
Adjusted R ²	0.2177	0.2735	0.2176	0.2731	
Panel B		Corruption control		Government effectiveness	
Parameter	(1)	(2)	(3)	(4)	
Intercept	−2.0451***	−2.1563***	−2.0500***	−2.1531***	
FLK	−0.0058***		−0.0056***		
FLK*Low Corruption control	−0.0046**				
FLK*Low Government effectiveness			−0.0105***		
FLJ		−0.1838***		−0.1815***	
FLJ*Low Corruption control		−0.0688			
FLJ*Low Government effectiveness				−0.0797***	
Low Corruption control	0.3317***	0.0324			
Low Government effectiveness			0.6140***	0.1005***	
Firm and country controls	Yes	Yes	Yes	Yes	
Fixed effects	Yes	Yes	Yes	Yes	
Observations	214,270	160,260	214,270	160,260	
Adjusted R ²	0.2176	0.2731	0.2177	0.2733	
Panel C		Regulatory quality		Anti-self-dealing	
Parameter	(1)	(2)	(3)	(4)	
Intercept	−2.0442***	−2.5142***	−2.0310***	−2.1483***	
FLK	−0.0058***		−0.0061***		
FLK*Low Regulatory quality	−0.0055***				
FLK*Low Anti-self-dealing			−0.0990		
FLJ		−0.1937***		−0.1871***	
FLJ*Low Regulatory quality		−0.0611***			
FLJ*Low Anti-self-dealing				−2.5224***	
Low Regulatory quality	0.3477***	0.0755***			

(Continues)

TABLE 6 | (Continued)

Panel C	Regulatory quality		Anti-self-dealing	
	(1)	(2)	(3)	(4)
Low Anti-self-dealing			6.3534	0.9548***
Firm and country controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
Observations	214,270	160,260	214,270	160,260
Adjusted R ²	0.2175	0.2731	0.2172	0.2728

Note: This table reports OLS estimates examining whether financial literacy (FL) constrains corporate cash holdings more strongly in countries with weak institutional governance. Weak governance is indicated by a dummy equal to 1 if the country's score on a given metric is below the sample median (and 0 otherwise). Panel A presents results for Rule of Law and Voice and Accountability; Panel B for Control of Corruption and Government Effectiveness; Panel C for Regulatory Quality and Anti-Self-Dealing Index (Djankov et al. 2008). The dependent variable is $CASH_{i,j,t}$, the natural logarithm of (cash and cash equivalents / total assets). Key interactions are $FL_{j,t-1} \times \text{Low Governance}$. All independent variables are lagged one year. Standard errors are robust and clustered at the firm level.

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

legal protection and enforcement quality in shaping corporate financial decisions. More critically, they establish societal FL as a functional substitute for formal governance mechanisms. Where institutional safeguards are deficient, FL empowers external stakeholders to monitor and discipline managerial behavior, thereby curbing agency-driven liquidity hoarding. This substitution effect provides compelling additional support for Hypothesis 1 and underscores the global governance relevance of financial sophistication beyond domestic institutional frameworks.

4.6 | Economic Significance of FL Versus Composite Country Governance

To facilitate direct comparison of the economic magnitudes of FL and formal country governance, we construct a composite governance index as the first principal component of the six institutional quality measures employed in this study. This approach captures the common variation across Control of Corruption, Government Effectiveness, Rule of Law, Voice and Accountability, Regulatory Quality, and the anti-self-dealing index (Djankov et al. 2008).

Results are reported in Table 7. Consistent with prior evidence, stronger country governance is associated with lower corporate cash holdings across all subsamples, though the effect is most pronounced in developed markets. The economic significance of both FL and the composite governance index is summarized in the bottom panel of the table. A one-standard-deviation increase in FLK reduces cash holdings by 8.9% of the sample mean in the full sample, 42.1% in developed economies, and 57.8% in developing economies. The corresponding figures for FLJ are 27.9%, 28.2%, and 28.8%, respectively. These magnitudes are directly comparable to those of the composite governance index, which range from 8.8% to 26.0% across the same subsamples. Thus, FL is not only statistically robust but economically on par with formal institutional quality in constraining corporate cash accumulation.

4.7 | Endogeneity

To address potential endogeneity concerns, such as reverse causality or omitted variables that might jointly influence FL and corporate cash policies, we implement a 2SLS-IV approach. Following Jia et al. (2024), we instrument country-level FL using secondary school enrollment rate, a historically determined driver of human capital accumulation that satisfies both relevance and exclusion restrictions.

Results are reported in Table 8. The first-stage regressions confirm instrument strength: Secondary enrollment is positive and highly significant in predicting both FLK and FLJ. In the second stage, the instrumented FL coefficients remain negative and statistically significant at the 1% level. These estimates imply that a one-standard-deviation increase in financial literacy reduces corporate cash holdings by 25.5% (using IV-FLK) to 29.2% (using IV-FLJ) relative to the sample mean, magnitudes that align closely with, and in some cases exceed, the OLS baseline. The consistency of the IV results with our main findings alleviates endogeneity concerns regarding the baseline model.

4.8 | Alternative Explanations

To rule out the possibility that the baseline result is affected by other considerations that are related to firm cash holdings, we control for refinancing risk (Harford et al. 2014) and the nature (horizon) of cash needs (Huang and Ritter 2021) in regression and re-examine the relationship between FL and corporate cash holdings.

Firms that may be denied refinancing are unlikely to commit capital to long-term projects. Thus, refinancing risk elevates corporate cash holdings. Following Harford et al. (2014), we use the fraction of a firm's short-term debt to total debt for the next 3 years as a proxy for refinancing risk.

TABLE 7 | Economic significance of financial literacy vs. country-level governance.

Parameter	Whole sample		Developed countries		Developing countries	
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	−2.0297***	−2.1223***	1.3501***	0.3762***	−2.8692***	−3.5706***
FLK	−0.0066***		−0.0385***		−0.0609*	
FLJ		−0.1916***		−0.1939***		−0.1985***
Composite country governance	−0.04313*	−0.1062**	−0.2672***	−1.0561***	−0.6930***	−0.6637***
Firm and country controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	214,270	160,260	157,938	677,758	84,802	60,912
Adjusted R^2	0.2173	0.2732	0.2280	0.2438	0.3533	0.3950
Economic impact of a one standard deviation increase in FLK on Cash/TA	−8.92%		−42.10%		−57.81%	
Economic impact of a one standard deviation increase in FLJ on Cash/TA FLJ		−27.95%		−28.22%		−26.80%
Economic impact of a one standard deviation increase in country governance on Cash/TA	−8.82%	−6.84%	−4.08%	−15.47%	−25.95%	−25.01%

Note: This table compares the economic impacts of financial literacy (FL) and a composite country governance index on corporate cash holdings across the full sample, developed countries, and developing countries. The composite governance index is the first principal component extracted via principal component analysis from the six institutional quality measures: Control of Corruption, Government Effectiveness, Rule of Law, Voice and Accountability, Regulatory Quality (World Bank), and the Anti-Self-Dealing Index (Djankov et al. 2008). The dependent variable is $CASH_{i,j,t}$, the natural logarithm of (cash and cash equivalents/total assets). All independent variables are lagged 1 year. Standard errors are robust and clustered at the firm level. The last three rows report the percentage change in cash holdings relative to the sample mean for a one-standard-deviation increase in each variable.

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

TABLE 8 | Endogeneity, 2SLS instrumental variable regressions.

Parameter	FLK		FLJ	
	Stage 1	Stage 2	Stage 1	Stage 2
Intercept	25.4238***	−2.2153***	1.0931***	−3.2082***
IV-FLK		−0.0207***		
IV-FLJ				−0.2018***
Secondary enrollment	0.09390***		0.0094***	
Firm and country controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
Observations	194,606	196,173	171,243	156,173
Adjusted R^2	0.9636	0.2418	0.7817	0.2572

Note: This table addresses potential endogeneity in the baseline relationship between financial literacy and corporate cash holdings using two-stage least squares (2SLS-IV). The instrument is the secondary school enrollment rate (Jia et al. 2024). The dependent variable is $CASH_{i,j,t}$, the natural logarithm of (cash and cash equivalents/total assets). Columns (1) and (2) report first- and second-stage results for FLK and Columns (3) and (4) report first and second-stage results for FLJ using instrumented FL. All independent variables are lagged 1 year. Standard errors are robust and clustered at the firm level.

*** denotes significance at the 1% level.

TABLE 9 | Alternative explanations, refinancing risk, and long-term cash needs.

Parameter	(1)	(2)	(3)	(4)
Intercept	−1.8903***	−1.9115***	−2.0308***	−2.1490***
FLK	−0.0058**		−0.0061***	
FLJ		−0.2145***		−0.1871***
Refinancing risk	0.2230***	0.2782***		
Long-term cash needs			0.3849***	0.3840***
Firm and country controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
Observations	214,270	160,260	214,270	160,260
Adjusted R^2	0.1972	0.2470	0.2172	0.2728

Note: This table examines whether the baseline relationship between financial literacy (FL) and corporate cash holdings persists in the face of refinancing risk or long-term cash needs. Refinancing risk is proxied by short-term debt maturing within 1 year scaled by total debt. Long-term cash needs are measured as the 3-year-ahead average of capital expenditures plus R&D expense, scaled by total assets. The dependent variable is $CASH_{i,j,t}$, the natural logarithm of (cash and cash equivalents/total assets). Standard errors are robust and clustered at the firm level.

***, and ** denote significance at the 1% and 5% levels, respectively.

TABLE 10 | Financial literacy and the market value of excess cash.

Parameter	(1)	(2)
Intercept	1.3855***	1.1745***
XCash	0.0910*	0.0187
FLK*XCash	−0.0016***	
FLJ*XCash		−0.0159***
FLK	−0.0064***	
FLJ		−0.0108
Country controls	Yes	Yes
Fixed effects	Yes	Yes
Observations	196,155	87,760
Adjusted R^2	0.4381	0.4593

Note: This table reports OLS estimates of Equation (2). The dependent variable is $MV_{i,j,t}$, defined as (market capitalization + total debt)/total assets. $XCash_{i,j,t-1}$ is excess cash, computed as the residual of the expected cash model. The key coefficients of interest are on the interactions $FLK_{j,t-1} \times XCash_{i,j,t-1}$ and $FLJ_{j,t-1} \times XCash_{i,j,t-1}$. Specifications follow Fama and French (1998) and include current and future changes ($t - 2$ to t , t to $t + 2$) in earnings, R&D, interest, dividends, and net assets, plus the country-level controls from Equation (1), industry, and year fixed effects. Standard errors are robust and clustered at the firm level.

***, and * denote significance at the 1% and 10% levels, respectively.

According to the funding-horizon model of Huang and Ritter (2021), firms' financial policies are affected by the characteristics of their cash needs. That is, firms with long-lived cash needs tend to hold more precautionary cash. For example, firms that have high R&D spending are prone to have high expected levels of

future cash needs and thus temporarily increase cash holdings substantially.

In Table 9, we report results that control for the two alternative drivers of corporate cash holdings. As expected, refinancing risk

and long-term cash needs are positively related to cash holdings, respectively. The coefficients of FLK and FLJ remain negative and significant at the five percent or one percent levels, supporting Hypothesis 1.

4.9 | FL and Market Value of Corporate Cash

Table 10 presents results from the valuation regression specified in Equation (2). The interaction terms between FL and excess cash are negative and highly significant: $FLK \times XCash = -0.0016$ ($p < 0.0005$) and $FLJ \times XCash = -0.0159$ ($p < 0.0002$). These coefficients indicate that higher societal FL substantially reduces the market value assigned to excess cash holdings. A one-standard-deviation increase in FLK lowers the valuation of excess cash by approximately 2.2%, while the corresponding effect for FLJ is 2.7%. This discount reflects investor skepticism about managerial discretion over liquid assets in high-FL environments, where stakeholders are better equipped to detect and deter value-destroying cash deployment. The findings align closely with Dittmar and Mahrt-Smith (2007), who document that strong internal governance diminishes the marginal value of cash by curbing expropriation risk. By demonstrating an analogous external governance role for societal FL, our results provide direct and robust support for Hypothesis 2: FL not only constrains the level of corporate cash but also disciplines its valuation in capital markets by mitigating agency-driven hoarding incentives.

5 | Conclusion

This study establishes financial literacy (FL) as a powerful informal governance mechanism that disciplines corporate cash holdings worldwide. By equipping stakeholders with the ability to process economic information and evaluate managerial intent, FL enables more effective detection and penalization of opportunistic liquidity accumulation. Firms operating in high-FL environments thus hold significantly less cash, all else equal, a prediction that extends canonical governance models (Opler et al. 1999; Dittmar et al. 2003) into the domain of societal knowledge and informal institutions.

Our core finding, a robust negative relationship between country-level FL and corporate cash holdings, survives an extensive battery of robustness tests, including alternative cash definitions, subsample splits (developed vs. developing markets), and 2SLS-IV estimation. The economic magnitudes are substantial: A one-standard-deviation increase in FL reduces cash holdings by 8%–29% of the sample mean, depending on the measure and specification. These effects are not only statistically significant but comparable to, or in developing economies, often larger than, those of a composite index of formal institutional quality encompassing rule of law, corruption control, government effectiveness, regulatory quality, voice and accountability, and investor protection.

We further demonstrate that FL substitutes for weak formal governance. In countries with institutional scores below the median, the marginal constraining effect of FL on cash holdings is significantly amplified. This substitution is particularly pronounced in developing markets, where legal and regulatory enforcement

is often limited. The policy implication is clear: Investing in financial education can yield governance dividends on par with institutional reform, offering a scalable, low-cost mechanism to enhance corporate discipline in emerging economies.

Extending the analysis to capital market pricing, we show that FL reduces the marginal market value of excess cash, a hallmark of agency risk (Dittmar and Mahrt-Smith 2007; Pinkowitz et al. 2006). A one-standard-deviation increase in FL is associated with a 2.2%–2.7% lower valuation of excess cash, reflecting investors' heightened ability to price managerial discretion. This result confirms that FL operates through a monitoring-and-pricing channel, complementing internal governance in mitigating the dark side of corporate liquidity.

Our findings contribute to a burgeoning literature on informal institutions in corporate finance and economics (Huang and Shang 2024). While social trust, media scrutiny, and civic capital shape behavior through norms and reputation, FL functions via cognitive empowerment: It transforms passive stakeholders into active monitors capable of interpreting financial statements, challenging cash deployment decisions, and imposing market-based sanctions. This knowledge-based mechanism is uniquely actionable, unlike cultural traits; FL can be directly enhanced through education and public policy.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study

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Appendix

Variable Definitions

FLK	Country-level static measure of financial literacy reported by Klapper and Lusardi (2020).
FLJ	Time-varying annual measure of financial literacy (Jia et al. 2024), which is the first principal component extracted from a principal component analysis of the following variables: public education, tertiary enrollment, nonlife insurance, and number of bank branches.
Ln(Cash/TA)	The logarithmic value of (cash and cash-equivalent assets) scaled by total assets.
Firm size	The logarithmic value of total assets.
Leverage	(Long-term debt plus debt in current liabilities) divided by book assets.
NWC/assets	Net working capital to total assets.
Market-to-book ratio	(Book value of assets minus the book value of equity plus the market value of equity) scaled by the book value of assets.
Capex/assets	Capital expenditures to total assets.
Industry sigma	Industry sigma is the mean of standard deviations of cash flow/assets over the previous 5 years for firms in the same industry as defined by two-digit SIC code.
Dividend dummy	A dummy variable equals 1 in years in which a firm pays a common dividend. Otherwise, the dummy equals 0.
Cash flow/assets	(Earnings after interest, dividends, and taxes but before depreciation) divided by book assets.
RD/sales	Research and development expenses divided by sales. R&D is set equal to 0 when the value is missing.
Acquisition activity	Acquisition expenditures divided by total assets.
Domestic credit/GDP (%)	Financial resources provided to the private sector by depository corporations.
GDP growth	Annual growth rate of GDP.
GDP/CAPITA	The natural logarithm of the country's GDP per capita (in constant USD).
Market capitalization/GDP	Stock market capitalization divided by GDP.
Trust	Societal trust measured by average response to a World Values Survey question that probes a respondent's propensity to trust others in society. 1 = trust, 2 = No.
Countries	Argentina, Australia, Brazil, Canada, Chile, China, Colombia, Egypt, France, Germany, Greece, Hungary, India, Indonesia, Italy, Japan, Jordan, Malaysia, Mexico, the Netherlands, Nigeria, Norway, Pakistan, Peru, Philippines, Poland, Russia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Thailand, Turkey, United Kingdom, United States, Vietnam